Carbon-Aware Server Replacement

Iris Uwizeyimana, Natalie Enright Jerger
Department of Electrical and Computer Engineering, University of Toronto
Toronto, Canada
iris.uwizeyimana@mail.utoronto.ca, natalie.enrightjerger@utoronto.ca

***Abstract*—Cloud computing contributes significantly to the global carbon emissions. Server management, including server replacement policies heavily influences the carbon footprint of cloud computing. While periodically replacing servers with newer hardware enhances energy efficiency, in turn reducing the operational carbon emissions from running servers, it also increases the embodied carbon footprint associated with server manufacturing. We propose an analytical model to determine the optimal server replacement frequency to reduce the overall datacenter carbon footprint, encompassing both embodied and operational carbon. Using our model, we perform a case study to analyze the impact prolonged server lifetime has on the net carbon footprint of a high performance computing cluster.**

I. Introduction

The Information and Communication Technology (ICT) sector is responsible for roughly 3.9% of global carbon emissions, with datacenters being a major contributor [8]. In 2022, datacenters accounted for 2% of the global electricity demand, equivalent to 460 TWh of energy consumption – a figure projected to exceed 1000 TWh by 2026 [5]. Given the average global carbon emission of 436 gCO_2/kWh in 2022 [14], datacenter operations generated $\sim$220 million tons of CO_2 annually, comparable to the carbon footprint of Egypt [15]. The net carbon footprint of datacenters is even higher, as these estimates do not account for all carbon emission sources.

Datacenter carbon emissions fall into two categories: operational and embodied. Operational carbon is emitted from running datacenter workloads on servers (e.g., 220 million tons CO_2 above). The embodied carbon is emitted during the manufacturing of servers. Embodied carbon is a big contributor to the net carbon cost of datacenter and is estimated to make up to $\sim$46% of the total datacenter carbon footprint [1]. Addressing both emission sources is critical for effective carbon reduction.

This work explores server replacement policies as a strategy for reducing the net carbon footprint. While extending server lifespans reduces embodied carbon emissions by decreasing hardware refresh frequency, doing so naively can inadvertently increase the total emissions as shorter refresh cycles leverage the better energy efficiency of newer hardware to lower operational carbon. To navigate this trade-off, we propose an analytical model that determines the optimal server replacement period by balancing embodied carbon costs against operational carbon savings. Our findings reveal that extending server lifespans has the potential to reduce the overall carbon footprint, but only when workload placement is carefully managed to mitigate the operational inefficiencies of older hardware.

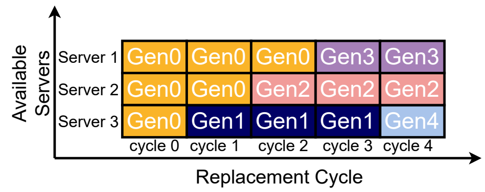

Fig. 1: Replacement Policy Illustration.

II. Cost Model

A. Server Replacement Policy

We assume a fixed server capacity in the datacenter, which means longer server lifespans reduce purchase frequency. Fig. 1 illustrates our cyclic replacement policy where servers in a cluster of S nodes are incrementally replaced over multiple cycles instead of a single bulk replacement. For example, with a 3-year replacement cycle and a 9-year replacement period, all servers are fully replaced after three cycles: $S/3$ nodes at the end of the first cycle, $2 \cdot S/3$ by the second cycle and all S nodes by the third. This policy prioritizes replacing the oldest servers first; the dark blue batch represents the first group replaced after retiring all the orange batches. Under this strategy, server batches achieve an average replacement period matching the 9-year operational timeframe. We adopt this policy to identify the optimal server replacement period for minimizing the net carbon footprint. This approach is realistic for cloud platforms, given the inherent hardware heterogeneity driven by asynchronous refresh cycles [2], [9], [12].

B. Calculating the Net Carbon Increase

We propose an analytical model to determine the optimal server batch replacement period, to minimize the net carbon footprint. When servers are replaced, the reduction in operational carbon ($OC_{reduction}$) is offset by increased embodied carbon ($EC_{increase}$), resulting in a net increase:

$$Net_{increase} = EC_{increase} - OC_{reduction}. \quad (1)$$

We derive expressions for $OC_{reduction}$ and $EC_{increase}$, assuming replacement period of X cycles for a cluster of S servers, with S/X servers replaced each cycle.

Operational Carbon Reduction ($OC_{reduction}$). Let OC_0 represent the original operational carbon emissions of S servers. Replacing S/X servers per cycle reduces the operational emissions due to improved energy efficiency. This average per-cycle reduction is characterized by a factor θ.

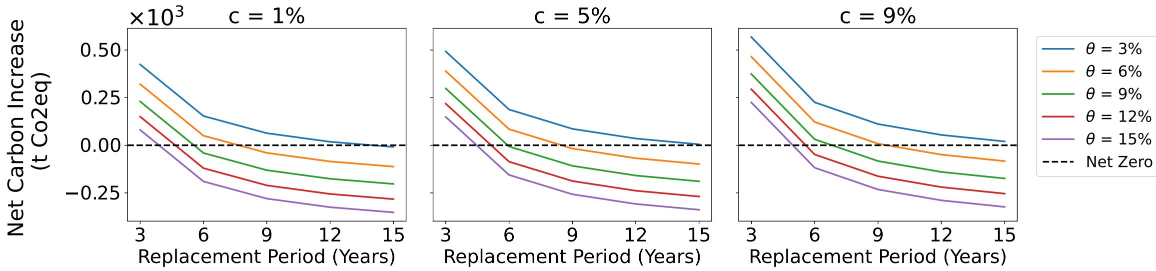

Fig. 2: Net Carbon Increase from Case study over 15 years (N=5 cycles).

Since the reduction in operational carbon footprint depends on the emissions from the previous cycle, we model $OC_{reduction}$ as a compounded reduction process, where:

- In the first cycle, the reduction in operational carbon compared to OC_0 is: $\Delta OC_1 = OC_0 \cdot \theta$
- In the second cycle, the reduction in operational carbon compared to OC_0 is: $\Delta OC_2 = OC_0 \cdot \theta \cdot (1-\theta)$ since $\Delta OC_2 = OC_1 \cdot \theta$ and $OC_1 = OC_0 - \Delta OC_1$.
- This can be extended to any cycle n where the reduction in operational carbon compared to OC_0 can be computed as $\Delta OC_n = OC_0 \cdot \theta \cdot (1-\theta)^{n-1}$

The total operational carbon reduction over N cycles is the cumulative sum, which simplifies to:

$$OC_{reduction} = OC_0 \cdot \left(1 - (1-\theta)^N\right) \tag{2}$$

Embodied Carbon Increase ($EC_{increase}$). We model the change in embodied carbon across server generations as a compounded factor c. For instance, if the initial embodied carbon footprint of a server is EC, the embodied carbon of the next generation server would be $EC \cdot (1+c)$, and for a server two generations newer, $EC \cdot (1+c)^2$. The factor c can be positive (indicating compounded growth) or negative (indicating compounded reduction). Since S/X servers are replaced every cycle with a new server generation, the increase in embodied carbon incurred at any cycle n can be computed as $\Delta EC_n = \frac{S}{X} \cdot EC \cdot (1+c)^n$. The total embodied carbon increase across all cycles can then be computed as the cumulative sum across all N cycles, which simplifies to:

$$EC_{increase} = \frac{EC \cdot S}{X} \cdot (1+c) \cdot \frac{1-(1+c)^N}{1-(1+c)} \tag{3}$$

Optimization Goal. Combining Equations 1, 2, and 3, the net carbon footprint increase over X cycles becomes:

$$\begin{aligned} Net_{increase} = &\left[\frac{EC \cdot S}{X} \cdot (1+c) \cdot \frac{1-(1+c)^N}{1-(1+c)}\right] \\ &- \left[OC_0 \cdot (1-(1-\theta)^N)\right] \end{aligned} \tag{4}$$

C. Case Study

To demonstrate our model, we analyzed Niagara, a high-performance computing cluster with 2,016 nodes, each containing 40 Intel Cascade or Skylake cores and 202 GB of RAM [10]. Niagara's average power consumption is approximately 900 kW per hour, translating to an annual operational carbon footprint of roughly 275,940 kg CO_2 based on Ontario's carbon intensity of 35 gCO_2/kWh [3]. We used a simplified version of the ACT model [6], by only incorporating the embodied carbon of DRAM, SoC components and packaging. This yielded an embodied carbon footprint estimate of 52.1 kg CO_2 per node. We set the server replacement cycle to three years, meaning $\frac{2{,}016}{X}$ nodes are replaced every three years over X cycles. The initial operational carbon OC_0 is estimated at 827.82 t CO_2.

Fig.2 shows that longer replacement periods consistently reduce total carbon emissions, although the marginal benefit diminishes with extended server lifespans. However, indefinitely extending server lifespan is impractical due to hardware degradations (e.g., HCI, NBTI [7], [11], [13]), increasing performance demands, software compatibility (such as modern instruction set requirements), and security risks from discontinued firmware support [4]. Therefore, rather than solely minimizing the carbon footprint, we focus on determining the server replacement period achieving a *net zero carbon increase*, where embodied carbon emissions match operational carbon savings. As illustrated in Fig. 2, the operational carbon reduction factor θ significantly affects desirable server lifespans. For example, at $c = 5\%$, a θ of 3% requires servers to last approximately 15 years (five cycles) to achieve net zero carbon increase. Increasing θ slightly to 6% shortens this requirement to about 9 years – a more feasible lifespan. Achieving higher θ values requires strategically leveraging heterogeneous server generations through careful workload-to-server placement and detailed workload characterization across diverse hardware.

III. Conclusion

We propose an analytical model to balance embodied and operational carbon emissions through optimal server replacement. Using the model, we demonstrate the impact operational carbon reduction rate – which is highly influenced by workload placement – has on minimizing the net carbon footprint while achieving feasible server lifespans. Consequently, we highlight the importance of scheduling techniques that leverages hardware and software heterogeneity to maximize operational carbon savings from new server deployments, thus promoting practical server lifespans that minimize the net carbon increase.

Acknowledgements

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